

Confusion matrices

Below are three confusion matrices for predictions with the dengue data.

Using a threshold of $c = 0.33$:

Predicted Status	Actual Status	
	negative	positive
negative	3069	430
positive	954	1267

Using a threshold of $c = 0.2$:

Predicted Status	Actual Status	
	negative	positive
negative	2327	239
positive	1696	1458

Using a threshold of $c = 0.5$:

Predicted Status	Actual Status	
	negative	positive
negative	3634	765
positive	389	932

Questions:

1. Which threshold would you choose?
2. How do accuracy, sensitivity, and specificity change when you change the threshold?